Katz - Messing

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§1. Review, statements

$X/\mathbb{F}_q$ smooth, projective, absolutely irred. var.

Recall. $N_m := \# X(\mathbb{F}_{q^m})$

$$Z(X/\mathbb{F}_q, T) = \exp\left(\sum_{m=1}^{\infty} \frac{N_m}{m} T^m\right)$$

$$T\frac{d}{dT} \log Z = \sum_{m=1}^{\infty} N_m T^m$$

Prop (Grothendieck)

$$N_m = \sum_{i=0}^{2n} (-1)^i \operatorname{Tr}(F \mid H^i_{ét}(X, \overline{\mathbb{Q}_\ell})) \quad , \quad (\ell \neq p)$$

Cor. $Z(X/\mathbb{F}_q) = \dfrac{P_0 \cdots P_{2n}}{P_1 \cdots P_{2n-1}}$, $P_i := \det(1 - TF \mid H^i(X, \overline{\mathbb{Q}_\ell}))$

Th. (Deligne) $P_i(T) = \prod (1 - \alpha_{ij} T)$ where $|\alpha_{ij}| = q^{i/2}$

Def. Let $\mathcal{H}$ be a cohomology theory; defined for $X/\mathbb{F}_q$ as above;

$\mathcal{H}$ is "reasonable" if

(1) PD: $\mathcal{H}^i(X) \times \mathcal{H}^{2n-i}(X) \longrightarrow \mathcal{H}^{2n}(X)$ perfect, $\dim \mathcal{H}^{2n} = 1$

& $F \mid \mathcal{H}^{2n}(X) = q^n$

(2) (Weak Lefschetz) $\exists\ d_0(X)$ s.t. $\forall d \geq d_0$, if $Y \subset X$ smooth hypersurface of deg d, $\mathcal{H}^i(X) \to \mathcal{H}^i(Y)$ is injective, $i \leq n-1$; surjective, $i \leq n-2$.

(3) define $\mathcal{P}^i = \det(1 - TF \mid \mathcal{H}^i)$

$$\longrightarrow \quad Z(X/\mathbb{F}_q) = \frac{\mathcal{P}^0 \cdots \mathcal{P}^{2n}}{\mathcal{P}^1 \cdots \mathcal{P}^{2n-1}}$$

Th. $\mathcal{H}$ is reasonable, $\longrightarrow$ $\mathcal{P}^i = P^i$.

Cor. (1) $\dim(H^i) = \dim(\mathcal{H}^i)$

(2) D1 - 3 hold for $\mathcal{H}$.

Deligne proved:

D1 : Riemann hypothesis

D3 : $L^i : H^{n-i}(X, \bar{\mathbb{Q}}_\ell) \xrightarrow{\sim} H^{n+i}(X, \bar{\mathbb{Q}}_\ell)$

(X_t) Lefschetz pencil on X

(WL). $H^{n-1}(X) \hookrightarrow H^{n-1}(X_t)$

$P^{n-1}(X) \mid P^{n-1}(X_t)$

$$\begin{array}{cc} (1-\alpha T) & f(T) \\ \mathrm{I} & \\ (1-\alpha^r T) & f(T)^{(r)} \end{array} \qquad \underset{t \in \mathbb{F}_{q^r}}{P^{n-1}(X)^{(r)} \mid P^{n-1}(X_t)}$$

D2: $\forall f(T)$, $f(T) \mid P^{n-1}(X)$

$\Leftrightarrow \forall t$ smooth, $f(T)^{(r)} \mid P^{n-1}(X_t)$

Th. $\mathcal{P}^i = P^i$.

Lemma. $X/\mathbb{F}_q$ enough to prove for $X/\mathbb{F}_{q^d}$.

Given Th_d, know RH $\mathcal{P}^i(d) = P^i(d)$ $\square$

Pf of Th induction on $n = \dim X$.

Pick (X_t) a rat'l Lefschetz pencil.

Weak Lefschetz, $i \leq n-2$, $\mathcal{H}^i(X) \xrightarrow{\sim} \mathcal{H}^i(X_t)$

$H^i(X) \xrightarrow{\sim} H^i(X_t)$

Interesting case : $i \in \{n-1, n, n+1\}$.

If t is any smooth parameter, $t \in \mathbb{F}_{q^r}$.

$$WL \longrightarrow \quad p^{n-1}(X/\mathbb{F}_q)^{(r)} \mid p^{n-1}(X_t/\mathbb{F}_q)$$

$$Z(X/\mathbb{F}_q) = \frac{P_0 \cdots}{P_1 \cdots} = \frac{p_0 \cdots}{p_1 \cdots}$$

We have $R^{n-1} \cdot R^{n+1} = R^n$, $\quad R^i = \dfrac{P^i}{p^i}$ is poly. (by D2)

Cor. (1) $\dim_k \mathcal{H}^i = \dim_{\overline{\mathbb{Q}}_\ell} H^i$.

(2) Conditions D1 - D3 also hold for $\mathcal{H}$.

Pf D1 = RH ✓ $\qquad$ D2 = formal from th.

D3 : Hard Lefschetz.

enough to show $(a, b) \longmapsto ab L^i$ is perfect.

fix $f : Y \to X$ inclusion of a sm. $d >> 0$ hypersurface

$f_*(1) = L$, and (therefore), we have

$$ab L^{n-i} = f_*\left(f^*(a) f^*(b) L^{n-i-1}\right)$$

Define $I = \operatorname{im}(f^* : H^i(X) \to H^i(Y))$. show $I \cap I^\perp = \{0\}$.

$$f_* : H^{2n-2}(Y) \xrightarrow{\sim} H^{2n}(X)$$

$$0 \longrightarrow I \cap I^\perp \longrightarrow I \oplus I^\perp \longrightarrow \mathcal{H}^{n-1}(Y) \longrightarrow \mathcal{H}^{n-1}(Y)/(I + I^\perp) \longrightarrow 0$$

Apply char. poly. $p^{n-1}(X/\mathbb{F}_q) = Z(I)\, Z(I^{\perp}/(I \cap I^{\perp}))\, Z\left(\dfrac{\mathcal{H}^{n-1}(Y)}{I + I^{\perp}}\right)$

Via PD. $(I \cap I^{\perp})^* = \mathcal{H}^{n-i}(Y)/(I + I^{\perp})$ [with q^n]

$$Z(I) = p^{n-1}(X/\mathbb{F}_q, T)^{(r)}.$$

$$p^{n-1}(Y/\mathbb{F}_q) = p^{n-1}(X/\mathbb{F}_q, T)^{(r)} [\ ].$$

By D2, Conclude $I \cap I^{\perp} = 0$. --- $\square$

Cor (Ogus) $\forall$ sm. hyperplane sections $Y \hookrightarrow X$, Letsihetz holds.

Pf $i \le n-2$, $\mathcal{H}^i = H^i$, thus $i \le n-1 \to$ enough to show injectivity.

But
$$\begin{tikzcd} \mathcal{H}^i(X) \arrow[r] \arrow[dr, "\sim"'] & \mathcal{H}^i(Y) \arrow[d] & \boxed{\text{injective}} \\ & \mathcal{H}^{2n-i}(X) & \end{tikzcd}$$

$[\Delta] \in \mathcal{H}^{2n}(X \times X) \xrightarrow{\text{Künneth}} \sum \alpha_i \otimes \beta_{2n-i}$, $\quad \alpha_i, \beta_i \in \mathcal{H}^i(X)$

Prop. α_i, β_i are $\mathbb{Q}$- algebraic cycles.

If $\sigma \in H^{2n}(X \times X)$, get $f_\sigma : \mathcal{H}^*(X) \to \mathcal{H}^*(X)$

p^i, p^i are pairwise coprime. CRT: $\exists\ \Pi^i(T) \in \mathbb{Q}[T]$ divis. by p^j, $j \neq i$, $\Pi^i \equiv 1 \mod p^i$

$\Pi^i(F) \curvearrowright \mathcal{H}^*$ kills $\mathcal{H}^j$, $j \neq i$ by Cayley-Hamilton.